Higgs Inflation
and
The Electroweak Gauge Sector

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Ending Inflation & The Hot Big Bang – SCGP 2023

Based on 2306.xxxxx w/ Stephon Alexander, Humberto Gilmer, and Katherine Freese



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Roadmap

I will…

I) Introduce a new method for the Higgs to be the inflaton (using EW gauge fields)

II) Demonstrate an approach to solving both the eta and hierarchy problem under the same guise.

III) Show that such constructions require high inflationary scales

IV) Hint to how warm constructions can avoid such constraints

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What is (Traditional) Higgs Inflation?

Unitary Gauge

$$H = (0, h/\sqrt{2})^{\mathsf{T}}$$

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu h \partial_\nu h - V(h) + \frac{M_{\rm pl}^2}{2} R + \frac{\xi}{2} h^2 R \right]$$

$$\hat{g}_{\mu\nu} = \Omega^2(h) g_{\mu\nu}, \quad \Omega(h) \equiv 1 + (h/\Lambda)^2$$

$$S = \int d^4x \sqrt{-\hat{g}} \left[-\frac{1}{2} \hat{g}^{\mu\nu} \partial_\mu \chi \partial_\nu \chi - \frac{V(\chi)}{\Omega^4(\chi)} + \frac{M_{\rm pl}^2}{2} \hat{R} + \frac{1}{2} \xi \chi^2 \hat{R} \right]$$

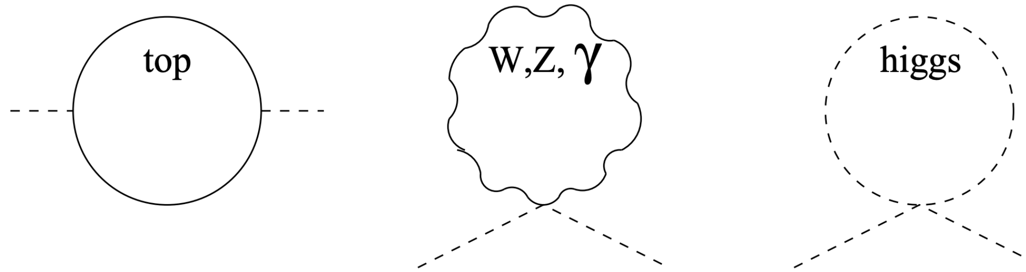
$$V(h) = \frac{\lambda}{4} \left(h^2 - v_{\rm EW}^2 \right)^2 \quad v_{\rm EW} \simeq 250 \text{ GeV}$$

Hierarchy and Eta Problems

Eta Problem

$$\frac{\Delta V}{(\Delta\phi)^4} \leq \mathcal{O}\left(10^{-6} - 10^{-8}\right)$$

Hierarchy Problem



$$\begin{array}{rrcl} \text{top loop} & -\frac{3}{8\pi^2}\lambda_t^2\Lambda^2 & \sim & -(2\ \text{TeV})^2 \\ SU(2)\ \text{gauge boson loops} & \frac{9}{64\pi^2}g^2\Lambda^2 & \sim & (700\ \text{GeV})^2 \\ \text{Higgs loop} & \frac{1}{16\pi^2}\lambda^2\Lambda^2 & \sim & (500\ \text{GeV})^2. \end{array}$$

$$\Lambda_{top} \lesssim 2\ \text{TeV} \qquad \Lambda_{gauge} \lesssim 5\ \text{TeV} \qquad \Lambda_{Higgs} \lesssim 10\ \text{TeV}$$

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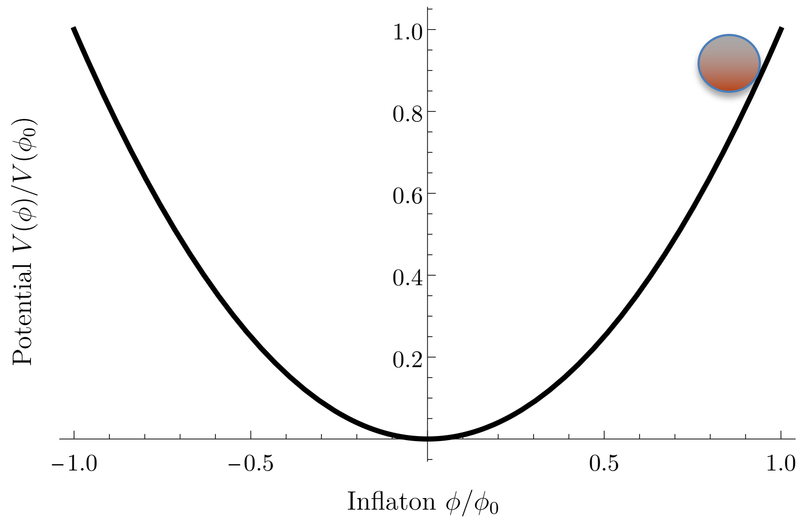
pNG Higgs Inflation

$$\mathcal{L} = D_\mu H^\dagger D^\mu H + V(H) + \frac{1}{4} W_a^{\mu\nu} W^a_{\mu\nu} + \frac{1}{4} B^{\mu\nu} B_{\mu\nu}$$
$$+ \frac{g^2 \beta^2}{2 f^2} \left(H^\dagger H\right) \mathrm{Tr}\left[W_a^{\mu\nu} \tilde{W}^a_{\mu\nu}\right]$$



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Lightning Review: (Non-Linear) Sigma Models & Pions

$$\mathcal{L} = \frac{F_\pi^2}{4}\partial_\mu \Sigma^\dagger \partial^\mu \Sigma \qquad SU_L(2) \times SU_R(2) \to SU(2)_{\text{isospin}}$$

$$\Sigma = (1 + \sigma/F_\pi)\exp\left(\frac{2i}{F_\pi}\Pi\right)$$

$$\Pi = \pi^a \tau^a = \begin{pmatrix} \pi^0 & \frac{1}{\sqrt{2}}\pi^- \\ \frac{1}{\sqrt{2}}\pi^+ & -\pi^0 \end{pmatrix}$$

$$\pi^0 = \pi^3,\ \pi^\pm = \frac{1}{\sqrt{2}}(\pi^1 \pm i\pi^2)$$

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The Littlest Higgs: Higgs as a pNG Boson

$$\mathcal{L}_{\text{NLSM}} = \frac{f^2}{8}\text{Tr}\left[\partial_\mu \Sigma^\dagger \partial^\mu \Sigma\right]$$

$$SU(5) \to SO(5)$$

$$\Sigma = \exp\left[\frac{2i}{f}\Pi\right]\Sigma_0$$

$$\Sigma_0 = \begin{pmatrix} & & \mathbb{1}_2 \\ & 1 & \\ \mathbb{1}_2 & & \end{pmatrix}$$

$$\Pi = \begin{pmatrix} \omega - \frac{\eta}{\sqrt{20}}\mathbb{1}_2 & \frac{H}{\sqrt{2}} & \phi^\dagger \\ \frac{H^\dagger}{\sqrt{2}} & \sqrt{\frac{4}{5}}\eta & \frac{H^\mathsf{T}}{\sqrt{2}} \\ \phi & \frac{H^*}{\sqrt{2}} & \omega^\mathsf{T} - \frac{\eta}{\sqrt{20}}\mathbb{1}_2 \end{pmatrix}$$

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The Littlest Higgs: Gauge Bosons and Fermions

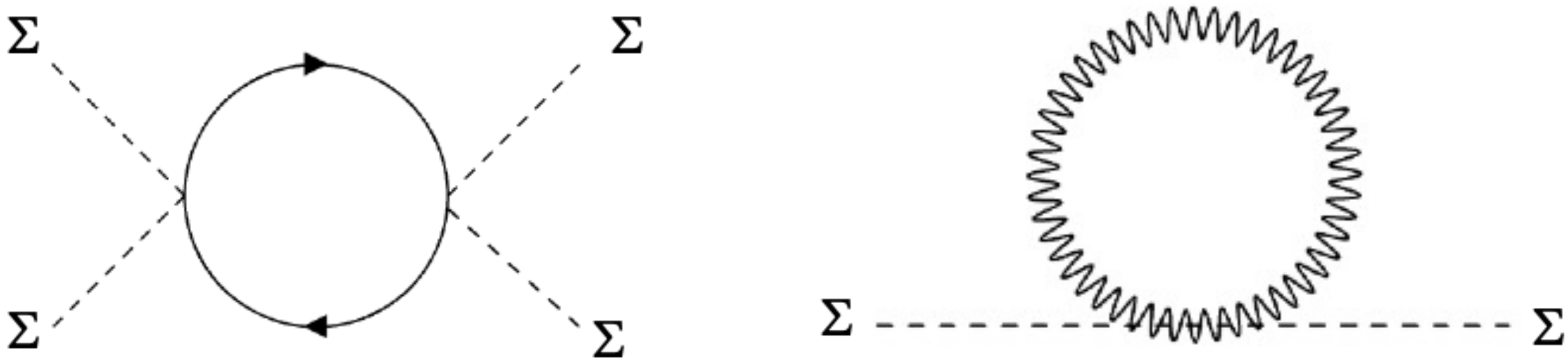
$$\text{Gauged}: SU(2) \times [SU(2) \times U(1)] \subset SU(5)$$

$$\mathcal{L}_{\text{NLSM}}^{\text{gauged}} = \frac{f^2}{8} \text{Tr}\left[(D_\mu \Sigma)^\dagger D^\mu \Sigma \right]$$

$$\mathcal{L}_{\text{top}} = -\frac{\lambda_1}{4} f \psi_i^\dagger \epsilon_{ijk} \epsilon_{xy} \Sigma_{jx} \Sigma_{ky} t^\dagger + \lambda_2 f T i\sigma_2 T + \text{H.c}$$

$$\psi_i = \begin{pmatrix} -i\sigma_2 q_3 \\ T \end{pmatrix} = \begin{pmatrix} b \\ t \\ T \end{pmatrix}$$

The Littlest Higgs: Coleman-Weinberg Potential



$$V_0(h) = D + f^4\left[cg'^2\sin^2\left(\frac{2h}{f}\right) + \frac{1}{2}\lambda_+\sin^4\left(\frac{h}{f}\right)\right]$$

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Beyond Littlest Higgs

$$V_0(h) = \mu^4 \left[c g'^2 \sin^2\left(\frac{2h}{f}\right) + \frac{1}{2}\lambda_+ \sin^4\left(\frac{h}{f}\right)\right]$$

$$\mathcal{L}_{\mathrm{CS}} = \sum_{j=1}^{2} \frac{g_j^2 \beta^2}{16} \mathrm{Tr}\left[Y\Sigma\,(Y\Sigma)^*\right] \mathrm{Tr}\left[W_{ja}^{\mu\nu} \tilde{W}_{\mu\nu}^{ja}\right]$$

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Beyond Standard Littlest Higgs Inflation

$$\mathcal{L} = \frac{f^2}{8}\mathrm{Tr}\left[(D_\mu\Sigma)^\dagger D^\mu\Sigma\right] - \frac{1}{4}\sum_{j=1}^{2} W^{\mu\nu}_{aj} W^{aj}_{\mu\nu}$$

$$- i\psi^\dagger \not{D}\psi - i\bar{u}_i^\dagger \not{D}\bar{u}^i + \mathrm{H.c.}$$

$$- \frac{\lambda_1}{4} f\psi_i^\dagger \epsilon_{ijk}\epsilon_{xy}\Sigma_{jx}\Sigma_{ky}\bar{t}^\dagger + \lambda_2 f T i\sigma_2 T + \mathrm{H.c}$$

$$+ \sum_{j=1}^{2} \frac{g_j^2\beta^2}{16}\mathrm{Tr}\left[Y\Sigma\,(Y\Sigma)^*\right]\mathrm{Tr}\left[W^{\mu\nu}_{ja}\tilde{W}^{ja}_{\mu\nu}\right].$$

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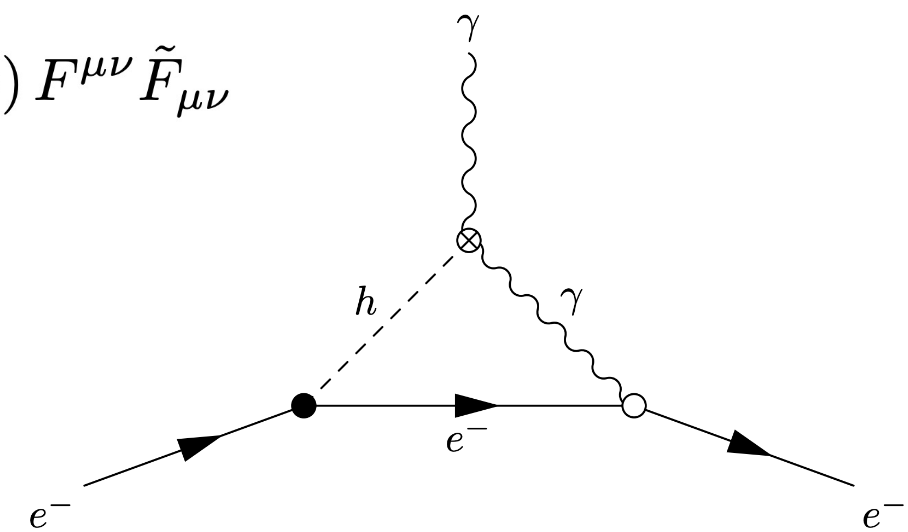
Electron Electric Dipole Moment Constraints

$$\frac{g^2\beta^2}{2f^2}\left(H^\dagger H\right)\left[W_a^{\mu\nu}\tilde{W}^a_{\mu\nu}\right] \supset \frac{e^2\beta^2}{4f^2}\left(hv\right)F^{\mu\nu}\tilde{F}_{\mu\nu}$$

$$H = [0, (v+h)/\sqrt{2}]^{\mathsf{T}}$$



ACME Collaboration:

$$d_e \lesssim 1.1 \times 10^{-29} e\ \mathrm{cm}$$

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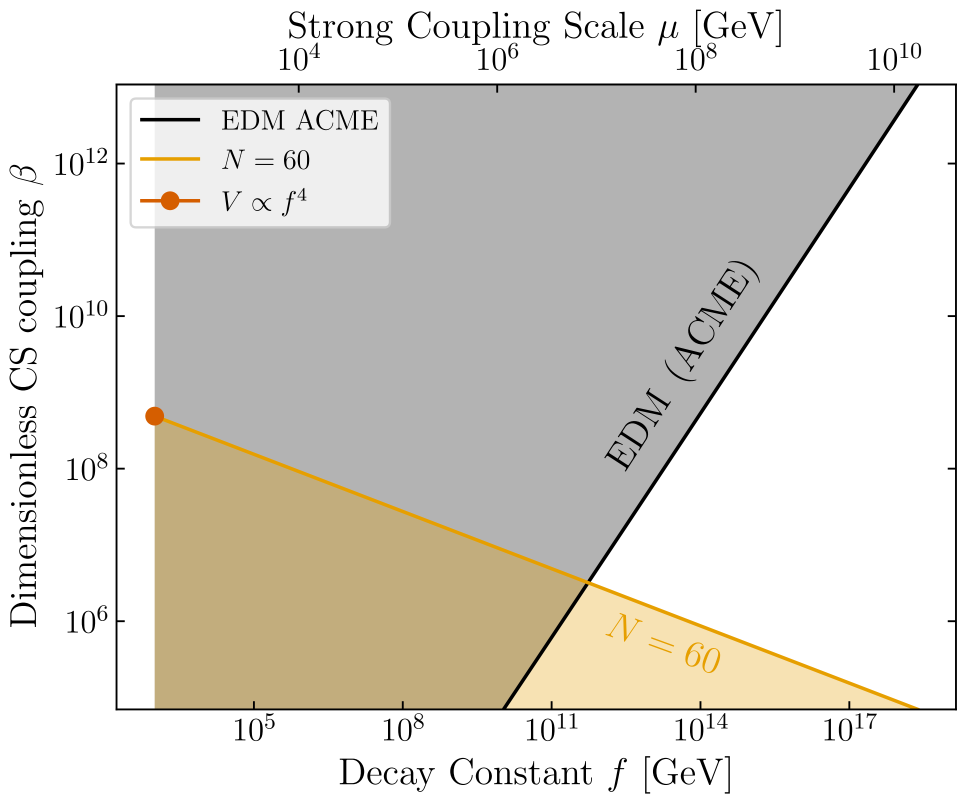
Parameter Space



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Conclusions/Future Extensions

1) New method for the Higgs to be the inflaton
2) Demonstrated a new approach to a simultaneous resolution of eta & hierarchy problems
3) While our cold constructions cannot do such a resolution, warm ones might.

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1) Warm Variety
2) Concrete UV Completion for amplitude
3) Systematic Inclusion of additional Fields
4) Two-point analysis of perturbations
5) Cosmological Collider Signals

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Extra Content

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Added Details

$$Q_1^a = \begin{pmatrix} \frac{\sigma_a}{2} & \\ & \end{pmatrix}$$

$$Q_2^a = \begin{pmatrix} & \\ & -\frac{\sigma_a^*}{2} \end{pmatrix}$$

$$Y = \frac{1}{2}\begin{pmatrix} \mathbb{1}_2 & \\ & \end{pmatrix}$$

$$D_\mu \Sigma = \partial_\mu \Sigma + i g_j W_\mu^{aj} \left(Q_a^j \Sigma + \Sigma Q_a^{j\mathsf{T}} \right) + i g' B_\mu \left(Y \Sigma + \Sigma Y \right)$$

$$\mathcal{L} \supset \lambda_t q_3 h \bar{t} \qquad \text{where} \qquad \lambda_t = \frac{\lambda_1 \lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

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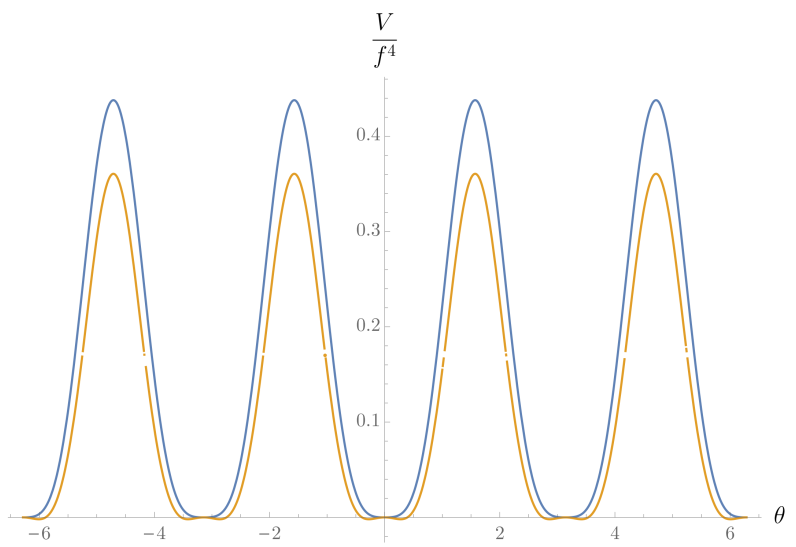
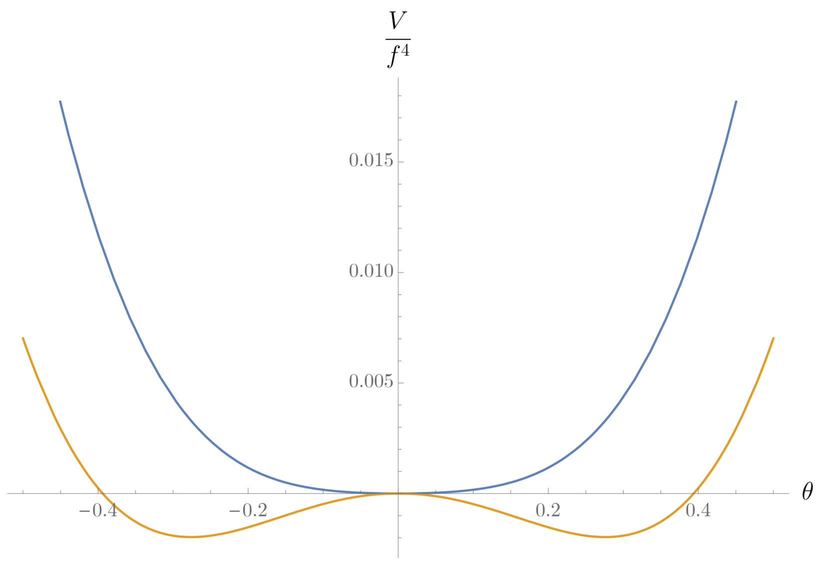
Electroweak Symmetry Breaking





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